Classical Field Theory — Exercise Sheet 7

Modes, complex pairs and admissibility

Questions only

Use $\{F,G\} = \int (F_\phi G_\pi - F_\pi G_\phi)$ and $\Omega = \int \delta\phi \wedge \delta\pi$. Except for the explicitly finite torus model in Problem 1, functionals have Lecture 7's smooth Schwartz-valued gradients. A formal distributional derivative does not automatically belong to that class.

Exercise 7.1 (Real oscillators in a finite torus model). On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, fix an integer $N \geq 2$ and use the real basis

$$e_0 = (2\pi)^{-1/2}, \qquad e_{n,c} = \pi^{-1/2}\cos(nx), \qquad e_{n,s} = \pi^{-1/2}\sin(nx), \quad 1 \leq n \leq N.$$

Let $\phi = q_0 e_0 + \sum_{n=1}^{N}(q_{n,c}e_{n,c} + q_{n,s}e_{n,s})$ and let π have the same expansion with coefficients p.

(a) Verify orthonormality and compute the pulled-back canonical form and the Hamiltonian $H = \frac{1}{2}\int_{\mathbb{T}}(\pi^2 + \phi_x^2 + m^2\phi^2)\mathrm{d}x$, where $m \geq 0$. Derive the oscillator equations and treat the zero mode at $m = 0$.

(b) Define $\widehat{\phi}_n = (2\pi)^{-1/2}\int e^{-inx}\phi\mathrm{d}x$, and similarly $\widehat{\pi}_n$, for $|n| \leq N$. Express them in the real coordinates and compute their complex bilinear brackets, including $n = 0$.

(c) Compute all nonzero real coefficients and the energy of the data $\phi = 3\cos(2x) - 2\sin x$, $\pi = 5\sin(2x) + \cos x$.

Exercise 7.2 (A complex scalar is two real canonical pairs). Write $\Phi = \phi_1 + i\phi_2$, with both real fields Schwartz, and take

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\nabla\Phi|^2 - m^2|\Phi|^2\big), \qquad m \geq 0.$$

Derive the two momenta and Hamiltonian by real variation. Compute all functional derivatives of $Q = \int(\phi_1\pi_2 - \phi_2\pi_1)\mathrm{d}^d x$ and its action on both fields and both momenta. Identify its finite flow, prove $\{Q,H\} = 0$, and verify admissibility in the product of the two real Schwartz phase spaces. Evaluate Q on $(\phi_1,\phi_2;\pi_1,\pi_2) = (f, 2f; -f, 3f)$, where $f \in \mathcal{S}(\mathbb{R}^d;\mathbb{R})$ and $\|f\|_2 = 1$.

Exercise 7.3 (A smooth window and a sharp interval). In one dimension let $w \in C_c^\infty(\mathbb{R};\mathbb{R})$ be a fixed window and set

$$F_w[\phi,\pi] = \frac{1}{2}\int_{\mathbb{R}} w(x)\phi_x(x)^2\mathrm{d}x, \qquad F_\#[\phi,\pi] = \frac{1}{2}\int_0^1 \phi_x(x)^2\mathrm{d}x.$$

Compute both derivatives against arbitrary real Schwartz variations. Prove F_w is admissible. Is $F_\#$ an admissible functional on the whole Schwartz phase space? Test the field $\phi(x) = (x + x^2)\eta(x)$, where $\eta \in C_c^\infty(\mathbb{R})$ equals one on a neighbourhood of $[0,1]$. Display its endpoint distribution terms.

Exercise 7.4 (The sign of the Hamiltonian commutator). For admissible F, G, let $H_F = D(\operatorname{grad} F)$ and $\mathbb{J}(a,b) = (b,-a)$ as in Lecture 7. Define the vector-field bracket by

$$[X,Y](z) = DY(z)X(z) - DX(z)Y(z),$$

the sign for which it acts on functionals as $X(YK) - Y(XK)$. Use the actual bracket-gradient formula of Lecture 7 to prove $X_{\{F,G\}} = -[X_F, X_G]$. For a nonzero example, choose real $f \in \mathcal{S}$ with $\|f\|_2 = 1$, put $q = \int f\phi$, $p = \int f\pi$, and take $F = q^2/2$, $G = p^2/2$. Compute both vector fields, their commutator, and $X_{\{F,G\}}$.